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CMAN

1.

Layer	Formula	# MAC Operations
Base	Nodes · $\frac{\text{Connections}}{\text{node}}$	N/A
1	784 nodes · $\frac{256 \text{ connections}}{\text{node}}$	200,704 MAC operations
2	256 nodes $\frac{128 \text{ con.}}{\text{node}}$	32,768 MAC operations
3	128 nodes $\frac{10 \text{ connections}}{\text{node}}$	1280 MAC operations

2. Total MACs for forward pass:

200,704 + 32,768 + 1280

= 234,752 MAC operations per forward pass

CamScanner

3. Now that we know our Total MAC Operations, we can determine the # of weights (trainable Parameters). Assuming one Parameter per operation,

234,752 trainable Parameters are utilized.

4. Given 234,752 weights and 4 Bytes Per weight, total memory is:

$$234{,}752 \text{ weights} \cdot \frac{4 \text{ bytes}}{\text{weight}} = 939{,}008 \text{ bytes}$$

5. 784 + 256 + 128 + 10 = 1178 "neurons"

and we need 1 activation per neuron, so:

1178 activations

$$1178 \text{ activations} \cdot \frac{4 \text{ bytes}}{\text{activation}} = 4712 \text{ bytes}$$

6. (2 x 234,752) / (4712 + 939,008)

469,504 / 943 720

= 0.498 FLOPS/byte

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